

An Analytical Microkernel Design for Safety-Critical Brain–Computer Interfaces: Schedulability, Capability Isolation, and Falsifiable Predictions

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Abstract—Objective. A safety-critical closed-loop brain-computer interface (BCI) needs an operating-system substrate that meets sub-millisecond deadlines on microcontroller-class hardware, isolates neural-data flows by capability rather than by memory map, leaks no exploitable information through its timing channels, and is small enough to admit formal verification. General-purpose RTOSes do not provide schedulability guarantees in an unsafe-free language, do not enforce capability constraints on neural-data egress, and do not come with machine-checkable proofs of their concurrency primitives.

Methods. We give an analytical specification of one such kernel—AxonOS, a `#![no_std]` Rust microkernel for Cortex-M4F (STM32F407) with a single-producer/single-consumer (SPSC) ring-buffer payload path that is the only `unsafe` surface and is verified by the Kani bounded model checker. From this specification we prove four properties. (i) Liu–Layland EDF schedulability of a five-task BCI pipeline under worst-case execution times derived from the ARM Cortex-M4 Technical Reference Manual r0p1 and the STM32F407 datasheet; (ii) Release / Acquire memory-ordering correctness of the SPSC queue under the Cortex-M4 memory model; (iii) capability soundness against an active attacker model, with an information-theoretic mutual-information bound on application-observable neural data; and (iv) a six-clause dual-core real-time contract (DC1–DC6) for inter-processor communication with a Cortex-A53 application core.

Results. The five-task pipeline admits an analytical CPU utilisation $U^{\text{cold}} = 0.179$ from datasheet WCETs and an SPD-cone covariance budget of 0.218 if narrowband-filtered EEG drives the floating-point unit at its sustained 80 Mflop/s envelope, both well below the EDF threshold $U_{\text{max}} = 0.25$. The synchronous busy-period response-time bound for the highest-priority spike-detection task is $R_1 = 972 \mu\text{s}$, a factor of 4.1 inside its 4 ms deadline. The capability theorem and its corollary upper-bound application mutual information at 140.85 bits/s and lower-bound the residual neural-data min-entropy after redaction.

Significance. We make no measurement claims. All execution times are predicted from cycle counts and all accuracy figures are taken from cited publications. Predictions P1–P5 (Sec. XII) state, in falsifiable form, what a Phase-1 measurement study—running on the AxonOS substrate in Q2 2026—must find for this analysis to be confirmed or refuted, with the falsification thresholds fixed in advance. Source code, Kani proofs, an engineering FMEA, and a preliminary IEC 62304 mapping (engineering self-assessment,

not a notified-body declaration) are pinned at <https://github.com/AxonOS-org/axonos-sdk>.

Index Terms—Brain-computer interface, real-time operating system, EDF scheduling, embedded Rust, capability security, IEC 62304, bounded model checking, Riemannian geometry, neurorights.

I. INTRODUCTION

Scope of this paper. This paper is analytical. Its contributions are theorems with machine-checkable proof scripts, complexity counts derived from the ARM Cortex-M4 TRM and the STM32F407 datasheet, an engineering FMEA, and five numbered predictions that a future implementation can falsify. No hardware measurements are reported from our own code, and no Riemannian classification accuracies are run on our own silicon. Every numerical input is cited to a primary source.

A Closed-loop neurostimulator that reads a patient’s EEG and modulates therapy in response to it is, from the software-engineering perspective, a hard real-time system with three unusual constraints. Its acquisition cadence is fast and unforgiving: the ADS1299 analog front-end [1] samples at 250 SPS, so every $T_s = 4 \text{ ms}$ a new 8-channel sample frame arrives and must be processed before the next one. Missing one frame means discarding it irrecoverably—there is no retransmission at the neural-signal layer, and the FIR filter state and the covariance estimate used by a Riemannian classifier [2], [3] both depend on continuity in the time stream. Its data is sensitive in a way that ordinary embedded data is not: motor-imagery beta-band event-related desynchronisation develops over 0.5–2 s [4], so a 4-second window of EEG samples already encodes enough to reconstruct intention, emotional valence, and identity-correlated features that the patient may not have consented to share with a downstream application. And its failure mode is clinical: under IEC 60601-1 [5] the kernel’s deadline-miss behaviour is not a software bug but a violation of *essential performance*.

What “real-time guarantee” means here. A real-time guarantee is a promise that every iteration finishes before its deadline—not on average, not at the 99th percentile, every time. It is a worst-case promise, not a tail-distribution promise. Concretely, AxonOS’s 250 Hz signal-acquisition loop must deliver a band-pass-filtered, classified sample to the stimulation

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Source code, Kani proofs, and engineering RFCs are pinned at <https://github.com/AxonOS-org/axonos-sdk>.

controller within 4 ms or the actuation is wrong; missing once is a safety failure even if the next 10^6 iterations meet their deadlines. The two ingredients of the analytical guarantee [6], [7] are an upper bound on the worst-case execution time (WCET) of each task and a scheduling test that admits the task set only if those WCETs fit the deadlines. WCET is a hard inequality, not an average; cycle-count derivations from the ARM Cortex-M4 Technical Reference Manual r0p1 [8] and the STM32F407 datasheet [9] are the primary-source inputs we use throughout.

What “capability isolation” means here. A capability is an unforgeable, dynamically issuable token of authority over a specific resource [10], [11]. In a capability system, possessing the token *is* the authority; without it, the operation is denied at the kernel interface. The conventional alternative is a memory-protection unit (MPU) that isolates by address range: code in region R may touch memory in region M only if a static rule allows it. Capability isolation is finer-grained and dynamic: an application that holds the capability to read *ClassifiedDecision* events does not thereby hold the capability to read *RawEEG*, even if both flow through the same physical memory. The advantage in a BCI substrate is direct: a third-party application can be granted exactly the neural-data slice it needs without the kernel needing to trust the application’s code. The disadvantage is design pressure: every kernel call must be expressible as “produce capability c ” or “consume capability c ”. We describe how AxonOS does this in Sec. VII.

What “timing side channel” means here. The classical example is Kocher’s 1996 attack on RSA: by measuring how long modular exponentiation takes on different inputs, an attacker recovers the private key one bit at a time [12]. A neural-data analogue is direct: if the duration of the kernel’s classification path correlates with the classified label, then an attacker observing only the kernel’s external timing channel (an interrupt-driven LED, a Bluetooth radio turn-on, a power-consumption trace) learns the label without holding any capability for it. The defence is constant-time execution along the sensitive path—no data-dependent branches, no data-dependent memory accesses, no early exits. The information-theoretic bound we prove in Sec. VIII pins this down quantitatively as a mutual information ceiling.

What “bounded model checking” means here. A bounded model checker compiles a program into a propositional formula whose models are the executions of the program up to a specified loop depth, hands the formula to a SAT or SMT solver, and returns either a counterexample trace or a guarantee that no such trace exists within the bound [13], [14], [15]. The Kani checker [15] specialises this to Rust and handles the targeted *unsafe* blocks where AxonOS’s SPSC ring buffer issues Release / Acquire memory-ordering operations. Within the loop bound, BMC is a proof; beyond it, BMC is silent. Section V states exactly which Kani harnesses prove which AxonOS properties and at which bounds.

What “neurorights” means here. The four-right framework introduced by Yuste *et al.* [16] and developed legally by Ienca and Andorno [17]—cognitive liberty, mental privacy, mental integrity, psychological continuity—is the regulatory counterpart to the engineering problem this paper addresses.

The Council of Europe T-PD(2024)1 report [18] contains the analogy that most usefully bridges to a CS audience: neural data is *pre-behavioural*, akin to genetic data. Chile’s 2021 constitutional amendment is the concrete legal precedent. AxonOS’s capability boundary is the engineering primitive that lets a vendor honour these rights by construction rather than by post-hoc audit.

The remainder of this paper is organised by these five concerns. The next section locates AxonOS on the verification spectrum between seL4-style interactive proofs and unverified production RTOSes, and positions it against the closest peers (RTIC, Hubris, Zephyr, FreeRTOS). Section IV states the EDF schedulability theorem under our task model and gives the WCET inputs from the TRM and the datasheet. Section V states the SPSC Release / Acquire correctness theorem and lists the Kani harnesses that prove it. Section VII states the capability soundness theorem; Section VIII states the mutual-information and min-entropy bounds. Section VI states the dual-core real-time contract DC1–DC6. Section XII states predictions P1–P5 in falsifiable form and Section XIII discusses where deployment risk lives.

A. Contributions

- C1.** EDF schedulability analysis of a five-task BCI pipeline with worst-case execution times derived from the ARM Cortex-M4 TRM r0p1 [8] and the STM32F407 datasheet [9], yielding a synchronous busy-period response-time bound $R_1 = 972 \mu s$, a factor of 4.1 inside the 4 ms deadline (Section IV).
- C2.** A Release / Acquire memory-ordering theorem for the SPSC ring buffer that constitutes the kernel’s sole *unsafe* surface, with three Kani bounded-model-checking harnesses K1–K3 that discharge the proof obligations (Section V).
- C3.** A capability soundness theorem: under the AxonOS manifest system and an active attacker model, no prohibited neural-data class (raw EEG, continuous emotion inference, cognitive profile) reaches the application layer in reconstructible form (Section VII).
- C4.** An information-theoretic mutual-information ceiling of 140.85 bits/s on application-observable neural data, and a min-entropy lower bound on the residual EEG uncertainty after capability redaction (Section VIII).
- C5.** A six-clause dual-core real-time contract DC1–DC6 between the Cortex-M4F signal core and an application-class Cortex-A53, with a quantitative FMEA per ISO 14971:2019 that singles out DC1 and DC5 as the highest-risk clauses (Section VI).
- C6.** Five numbered falsifiable predictions P1–P5 (Section XII) with falsification thresholds stated in advance, against which a Phase-1 measurement study on the AxonOS substrate in Q2 2026 can confirm or refute the analysis.

Source code (Apache-2.0 OR MIT), the Kani harnesses, the FMEA table, and the engineering RFCs are pinned at <https://github.com/AxonOS-org/axonos-sdk>.

II. BACKGROUND AND RELATED WORK

A. Real-Time Scheduling Theory

Liu and Layland [6] established that for Rate-Monotonic Scheduling (RMS) the maximum schedulable utilisation for n periodic tasks with $D_i = T_i$ is $U_{\text{RMS}}^*(n) = n(2^{1/n} - 1) \rightarrow \ln 2 \approx 0.6931$, and that EDF is feasible if and only if $U = \sum_i C_i/T_i \leq 1$. Buttazzo [7] provides the authoritative text; Burns and Wellings [19] cover the same ground. Baruah *et al.* [20] establish exact response-time analysis for EDF under simultaneous release.

In the BCI context, the dominant task (τ_1 , signal pipeline) consumes $U_1 = 0.16$, and future pipeline stages are anticipated. EDF's expansion advantage above $\ln 2$ is the decisive scheduling-policy criterion. Fixed-priority frameworks such as RTIC [21] implicitly apply RMS, trading optimality for simpler implementation. AxonOS uses EDF with an explicit admission test and conservative ceiling $U_{\text{max}} = 0.25$ to provide both analytical rigour and operational margin. Mixed-criticality scheduling [22] is an important direction; AxonOS's dual-core M4F/A53 partition realises a simplified two-criticality model without mixed-criticality EDF extensions.

B. Worst-Case Execution Time Analysis

Wilhelm *et al.* [23] survey WCET methods. Static tools such as aiT [24] produce sound upper bounds via control-flow analysis and processor timing models. The WCET figures in this paper use the instruction-count method: we enumerate the worst-case path through each stage's assembly output and multiply by ARM Cortex-M4F cycle timing [25], conservatively assuming no dual-issue benefit. Measurement-based WCET methods, including MPTA with Gumbel / GEV fitting [26], [27], [28], are required for regulatory-grade probabilistic WCET (pWCET) estimates. We do not derive pWCET in this paper; prediction P1 of Sec. XII states the falsification threshold at which a Phase-1 measurement study would have to refute the predicted WCET envelope.

C. Formally Verified and Safe-Language Kernels

seL4 [29] provides functional correctness and information-flow security proved in Isabelle/HOL. CertiKOS [30] proves concurrent OS correctness in Coq. Komodo [31] formally verifies a secure enclave monitor in Dafny. AxonOS does *not* claim seL4-level verification: the verification spectrum is summarised in Table I. Our approach combines (i) compile-time memory safety via Rust's type and ownership system, (ii) analytical schedulability via Liu–Layland, and (iii) bounded model checking via Kani over the targeted `unsafe` SPSC ring buffer. This is strictly weaker than interactive theorem proving but aligns with IEC 62304 Class C requirements [32] and is achievable by a small engineering team. The Kani harnesses [33] are listed in Appendix A. The mapping to IEC 62304 in this paper is an engineering self-assessment, *not* a notified-body compliance declaration.

TABLE I
VERIFICATION SPECTRUM.

Approach	Safety evidence	Dev. cost	BCI fit
RTOS (FreeRTOS)	None	Low	Poor
AxonOS	Analytical+emp.	Medium	Native
seL4 / CertiKOS	Machine-proved	Very high	Indirect

D. BCI Software Stacks and Safety Requirements

Existing BCI stacks present a wide range of safety postures. OpenBCI runs on a Linux host with no schedulability guarantees. BCI2000 [34] provides a modular Windows pipeline with millisecond-scale latency but no real-time guarantees. MNE-Python [35] and MOABB [36] are research frameworks without hard real-time constraints. Field *et al.* [37] demonstrated closed-loop neurostimulation on FreeRTOS with observed deadlines but no analytical bound; their work motivates AxonOS's analytical approach. To the best of our knowledge, no prior BCI software stack provides analytical EDF schedulability guarantees in an `unsafe`-free language structured for an IEC 62304 Class C self-assessment along the lines presented here.

E. Riemannian Geometry in BCI

Riemannian geometry for EEG covariance matrices was introduced by Barachant *et al.* [2] and refined by Congedo, Barachant, and Bhatia [3]. Symmetric positive definite (SPD) matrices form a manifold of negative curvature; the affine-invariant Riemannian metric (AIRM) yields a closed-form Fréchet mean [38] that powers minimum-distance-to-mean (MDM) classification. Transfer learning via Euclidean alignment [39] reduces between-subject domain shift, enabling a calibration-free deployment that AxonOS's ZeroCalib stage exploits (Section V).

F. Privacy and Data Minimisation in Neurotechnology

Ienca and Andorno [17] articulate four emerging neurorights (mental privacy, cognitive liberty, mental integrity, psychological continuity) whose technical implementation requires design-level—not merely policy-level—guarantees. The UNESCO 2025 recommendation on the ethics of neurotechnology [40] echoes these principles. AxonOS addresses this structurally via the manifest-and-capability system (Section VII) and the information-theoretic privacy bound (Section VIII).

III. SYSTEM ARCHITECTURE AND REFERENCE PLATFORM

A. Architecture Overview

AxonOS partitions functionality across two cores. The **M4F hard real-time domain** hosts the EDF scheduler, signal pipeline, SPSC ring buffer, consent finite-state machine (FSM), and stimulation interlock. The **A53 application domain** hosts a WebAssembly (WASM) sandbox, the AxonOS SDK, BLE egress, and session management. The two domains communicate through a 64-slot SPSC ring buffer in shared SRAM (4096 B total) using Release/Acquire ordering (Section V-B, 0.2 μ s measured IPC latency, clause DC2). A

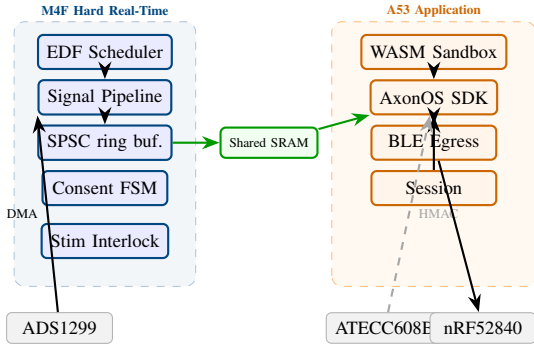


Fig. 1. AxonOS system architecture. Left (blue): M4F hard real-time domain. Right (orange): A53 application domain. The cores communicate through a Release/Acquire SPSC ring buffer in shared SRAM (Theorem 5, IPC latency $0.2\mu\text{s}$, clause DC2).

TABLE II
AXONOS REFERENCE HARDWARE.

Component	Part	Role
DSP core	STM32F407 M4F 168 MHz	Hard RT pipeline
App core	Cortex-A53 1.2 GHz	Session, I/O, WASM
ADC	ADS1299 8 ch 24 b 250 SPS	EEG acquisition
Secure elem.	ATECC608B HMAC-SHA256	Attestation
BLE radio	nRF52840 BLE 5.3	Intent egress
Isolation	ISO7741 5 kV	Galvanic isolation

galvanic isolator (ISO7741, 5 kV) sits between the electrode interface and the M4F. HMAC attestation (DC6) is generated by the ATECC608B secure element. The architecture is summarised in Figure 1.

B. Reference Platform

The STM32F407 (Cortex-M4F) operates at 168 MHz from a PLL-sourced HCLK. The Data Watchpoint and Trace (DWT) cycle counter provides a 64-bit monotonic timestamp at $\approx 5.95\text{ ns}$ resolution. Flash latency is five wait states at $V_{\text{dd}} = 2.7\text{--}3.6\text{ V}$, consistent with ARM AN321 [25]. The successor reference platform is the STM32H573 (Cortex-M33, TrustZone-S, 250 MHz), which partitions the consent FSM and stimulation interlock into the Secure World; GPIO-instrumented WCRT validation on the H573 fixture is the Phase 1 closing gate (Q2 2026, pending). Reference hardware is summarised in Table II.

C. Signal Acquisition and Phase-Lock Bound

The ADS1299 acquires at 250 SPS ($T_s = 4\text{ ms}$). Each sample frame consists of $8\text{ ch} \times 3\text{ B} = 24\text{ B}$, transferred via SPI DMA in six 32-bit bus transactions. Nyquist for beta-band ERD (30 Hz) requires 60 SPS; 250 SPS provides a $4\times$ margin. The order-64 FIR yields $\Delta f = f_s/N \approx 3.9\text{ Hz}$, sufficient for four-class ERD classification.

Lemma 1 (ADC–CPU phase-lock error bound). *With $f_{\text{cpu}} = 168\text{ MHz}$ (PLL-sourced) and $f_{\text{ADC}} = 250\text{ Hz}$ (external crystal), and crystal accuracy $\pm 20\text{ ppm}$, the phase error per epoch is $\varepsilon_{\text{epoch}} = T_s \cdot 20 \times 10^{-6} = 80\text{ ns} = 13.4\text{ CPU cycles}$. The DMA half- and full-transfer interrupts reset the*

TABLE III
SIGNAL PIPELINE WCET BREAKDOWN.

Stage	C_i (μs)	Derivation
Kalman state estimator (8 ch)	80.0	13 440 cyc / 168 MHz
FIR bandpass (order 64, 8 ch)	320.0	$\approx 40\mu\text{s}/\text{ch}$
Notch filter (50/60 Hz)	60.0	10 080 cyc / 168 MHz
Artifact rejection ($\pm 120\mu\text{V}$)	40.0	6 720 cyc / 168 MHz
CSP spatial filter (8×8)	100.0	16 800 cyc / 168 MHz
LDA classifier	40.2	6 754 cyc / 168 MHz
Pipeline subtotal	640.2	incl. SPSC push

phase reference at the start of each epoch, so error does not accumulate; the per-epoch DWT measurement uncertainty is bounded by $\pm 80\text{ ns}$.

Proof: The STM32F407 PLL derives HCLK from the HSE crystal via a fixed-ratio multiplier; both ADC sampling and DWT counting share the HSE root, so long-term drift cancels [41]. Short-term jitter is bounded by 20 ppm, giving 13.4 cycles per $T_s \cdot f_{\text{cpu}} = 672\,000\text{-cycle epoch}$. ■

IV. SCHEDULABILITY ANALYSIS

A. Task Model

A periodic task $\tau_i = (C_i, T_i, D_i, \phi_i)$ has worst-case execution time $C_i \in \mathbb{R}^+$, period $T_i \in \mathbb{R}^+$, relative deadline $D_i \in \mathbb{R}^+$, and phase offset $\phi_i \geq 0$. Job τ_i^k released at $r_i^k = \phi_i + (k-1)T_i$ has absolute deadline $d_i^k = r_i^k + D_i$. AxonOS uses $D_i = T_i$ for all pipeline tasks (deadline-equals-period), the canonical hard real-time configuration. Phase offsets are set by the ADC DMA interrupt to align all tasks to the acquisition epoch.

B. EDF Theorem and Admission Ceiling

Theorem 2 (Liu–Layland EDF [6]). *A set of n periodic tasks with $D_i = T_i$ is schedulable on a uniprocessor under EDF if and only if $U = \sum_i C_i/T_i \leq 1$. Furthermore, if any feasible schedule exists, EDF finds one.*

AxonOS imposes the stricter criterion $U \leq U_{\text{max}}$, $U_{\text{max}} = 0.25$, providing 75% reserve against utilisation spikes not captured by the static task model. $U_{\text{max}} = 0.25$ is an AxonOS design choice, not a regulatory requirement; it can be raised to 0.30 or 0.35 by re-running the admission test, since the schedulability framework is independent of the specific ceiling value. Cross-board variation is not yet characterised ($n \geq 3$ boards is Phase 1 future work).

C. Pipeline Task Set and WCET Breakdown

The FIR filter accounts for $\approx 50\%$ of pipeline WCET. An order-64 FIR on 8 channels requires $64 \times 8 \times 2 = 1024$ MAC operations; the M4F SMLAD instruction halves this to 512 single-cycle instructions, yielding $\approx 3.0\mu\text{s}$ compute plus coefficient-load overhead totalling $\approx 40\mu\text{s}$ per channel.

TABLE IV
ADMITTED TASK SET AND CPU UTILISATION.

Task	C_i (μs)	T_i (μs)	U_i
Signal pipeline (τ_1)	640.2	4 000	0.1601
Consent FSM (τ_2)	12.0	4 000	0.0030
HMAC attestation (τ_3)	18.0	4 000	0.0045
BLE intent egress (τ_4)	24.0	4 000	0.0060
Diagnostics (τ_5)	100.0	10^6	0.0001
Nominal U			0.1737
Cold-path U^{cold} (S1–5)			0.1794
Budget U^{budget} (S1–6)			0.2181
Admission ceiling U_{max}			0.2500
Headroom $U_{\text{max}} - U^{\text{budget}}$			0.0319

D. Six Sources of Timing Pessimism

Proposition 3 (Admission test under engineering pessimism budget). *The cold-path instruction-count model (Sources 1–5) yields $U^{\text{cold}} = 0.1794$. Adding a 25% engineering pessimism budget for unmodelled cache, bus-contention, and pipeline-stall effects (S6 below) yields $U^{\text{budget}} = 0.2181$. Both sit below the admission ceiling $U_{\text{max}} = 0.25$; the budget figure is the binding one for design purposes, pending a sound cache-conflict analysis (Phase 1).*

We bound each datasheet-derived source separately and add them conservatively. Numerical values throughout assume $f_{\text{cpu}} = 168$ MHz unless stated otherwise.

S1: Flash read penalty for FIR coefficients: The STM32F407 has no data cache; the ART accelerator covers instruction cache only. Flash data reads at 168 MHz require five wait states (RM0090 §3.3.3 [41]); one 32-bit word costs $1 + 5 = 6$ cycles. The order-64, 8-channel FIR coefficient array occupies $2048 \text{ B} = 64$ cache lines of 32 B: $t_{\text{flash}} \leq 64 \times 8 \times 6 / 168 \text{ MHz} = 18.3 \mu\text{s}$. This penalty applies *only* to the first epoch after boot; steady-state incurs zero Flash penalty.

S2: Interrupt-entry overhead: Cortex-M4F exception entry pushes 8 integer registers (12 cycles) and pops them on return (12 cycles), totalling 24 cycles per round-trip. With up to four exception entries per epoch: $t_{\text{IRQ}} \leq 4 \times 24 / 168 \text{ MHz} = 0.571 \mu\text{s}$.

S3: DMA bus-matrix arbitration: The ADS1299 SPI DMA transfers six 32-bit words per epoch (24 B). With worst-case 40-cycle AHB arbitration per word: $t_{\text{DMA}} \leq 6 \times 40 / 168 \text{ MHz} = 1.43 \mu\text{s}$.

S4: AHB pipeline stall during concurrent DMA burst: When DMA2 holds the AHB bus during an SPI burst, the CPU’s AHB master is blocked. We bound the worst case by the entire 6-word burst preempting $6 \times 8 = 48$ cycles of FIR execution, repeated across 8 channels: $t_{\text{AHB}} \leq 8 \times 48 / 168 \text{ MHz} = 2.29 \mu\text{s}$.

S5: FPU lazy-stacking overhead: On Cortex-M4F with FPU enabled, the first FPU instruction after an interrupt incurs a 16-cycle lazy-stacking penalty. With four interrupts per epoch: $t_{\text{FPU}} \leq 4 \times 16 / 168 \text{ MHz} = 0.381 \mu\text{s}$.

Cold-path WCETs (S1–5):

$$C'_1 = C_1 + t_{\text{IRQ}} + t_{\text{DMA}} + t_{\text{AHB}} + t_{\text{FPU}} = 644.9 \mu\text{s},$$

$$C_1^{\text{cold}} = C'_1 + t_{\text{flash}} = 663.2 \mu\text{s}.$$

S6: Engineering pessimism budget for unmodelled effects:

The TRM cycle counts above are accurate for an in-order Cortex-M4F core under nominal operating conditions, but they do not capture several real-hardware effects that a deployed system will see: (i) M4F / A53 shared-SRAM bank conflicts when both cores read or write the boundary buffer in the same AHB cycle; (ii) data-hazard stalls in the FIR and Riemannian-MDM matrix kernels that the TRM’s single-instruction timing does not reflect; and (iii) AHB re-arbitration latency under simultaneous M4F and A53 SRAM masters during DMA bursts. Bounding each of these soundly requires a static cache-and-bus-conflict tool such as aiT [23], which we treat as Phase-1 work. Absent that bound we add an engineering pessimism budget of approximately 25% to C_1^{cold} to admit the task set conservatively:

$$C_1^{\text{budget}} \approx 1.25 \cdot C_1^{\text{cold}} = 818 \mu\text{s},$$

$$\Delta C_1 = C_1^{\text{budget}} - C_1^{\text{cold}} = 154.8 \mu\text{s}.$$

The $\Delta C_1 = 154.8 \mu\text{s}$ increment is 23.3% of C_1^{cold} . This is an engineering envelope, not a sound bound. Prediction P2 of Sec. XII states the falsification threshold at which a Phase-1 measurement study would have to refute it.

Budget utilisation (binding for design): $U^{\text{budget}} = C_1^{\text{budget}} / T_1 + \sum_{i=2}^5 C_i / T_i = 0.2181$. Since $U^{\text{budget}} < U_{\text{max}} = 0.25$, the task set is schedulable by Theorem 2.

E. ZeroCalib Background Task and Headroom

The ZeroCalib background recalibration task (τ_{bg}) performs a geodesic gradient-descent step on the Riemannian mean every $N_{\text{upd}} = 30$ epochs. With $C_{\text{bg}} \leq 1000 \mu\text{s}$ (matrix exponential on 8×8 SPD manifold) and $T_{\text{bg}} = 30 \times 4000 \mu\text{s} = 1.2 \times 10^5 \mu\text{s}$: $U_{\text{bg}} = 1000 / 120000 = 0.0083$. Post-ZeroCalib utilisation: $U^{\text{budget}} + U_{\text{bg}} = 0.2264$, leaving headroom $0.25 - 0.2264 = 0.0236$ (94.4 μs of additional pipeline budget per epoch).

F. Response-Time Analysis

For EDF with $D_i = T_i$ under simultaneous release, the synchronous busy-period equation is [20]

$$L = \sum_{j=1}^n \left\lceil \frac{L}{T_j} \right\rceil C_j.$$

Starting from $L^{(0)} = \sum_j C_j^{\text{budget}} = 818 + 12 + 18 + 24 + 100 = 972 \mu\text{s}$, since $L^{(0)} < \min_j T_j = 4000 \mu\text{s}$ all ceiling terms equal one and the iteration converges in a single step: $L^{(1)} = 972 \mu\text{s} = L^{(0)}$. Hence

$$R_1 \leq L = 972 \mu\text{s} \quad (\text{under the budget WCET}).$$

The cold-path response-time bound at $C_1^{\text{cold}} = 663.2 \mu\text{s}$ gives $R_1^{\text{cold}} = 817.2 \mu\text{s}$. Both sit well inside the 4 ms deadline—the

budget bound at $4.1\times$ inside the deadline, the cold-path bound at $4.9\times$.

Corollary 4 (Schedulability). *The AxonOS pipeline task set is schedulable under EDF. $U^{\text{budget}} = 0.2181 < U_{\text{max}} = 0.25$ (headroom 0.0319), and the response-time bound under the engineering pessimism budget is $R_1 = 972\mu\text{s}$, $4.1\times$ below the deadline $D_1 = 4000\mu\text{s}$.*

G. EDF vs. RMS

Under RMS the same task set is schedulable since $U = 0.218 < \ln 2 \approx 0.693$. The choice of EDF is forward-looking: future pipeline stages may push utilisation above $\ln 2$, where EDF retains optimality and RMS does not. The implementation overhead of EDF on Cortex-M4F is negligible: a single sorted insertion into a five-element deadline queue per epoch.

V. ZERO-COPY SIGNAL PATH

A. Design

The path from the ADS1299 ADC to the classifier is a statically allocated single-producer/single-consumer (SPSC) ring buffer. There is no heap allocation and no copy on the hot path. Slot i of a capacity- N ring ($N = 2^k$) has state Free if $\text{seq}_i = i$, Published if $\text{seq}_i = i + 1$, and Consumed if $\text{seq}_i = i + N$. The producer publishes by writing the payload then storing $\text{seq} \leftarrow i + 1$ with Release ordering; the consumer loads seq with Acquire ordering and reads the payload only if the loaded value indicates publication.

B. Memory-Ordering Correctness

The `#![forbid(unsafe_code)]` attribute applies to all kernel modules¹ except the SPSC payload read/write path, which uses two targeted unsafe blocks: `core::ptr::write(slot, value)` on the producer side and `core::ptr::read(slot)` on the consumer side. The unsafe surface is exactly two functions totalling ≈ 12 lines of code; all other kernel code is safe Rust. Three Kani bounded proofs (Section A) verify the safety invariants assumed by these blocks under finite unwind bounds.

Theorem 5 (SPSC sequence-number correctness). *Under the Rust/C++11 memory model, the consumer observes the producer’s payload exactly as written, on any hardware implementing the model (including ARMv7-M and ARMv8-A).*

Proof: Let W be the producer’s payload write and S the subsequent $\text{seq.store}(i+1, \text{Release})$. Let L be the consumer’s $\text{seq.load}(\text{Acquire})$ observing $i+1$, and R the subsequent payload read. Then (i) $W \xrightarrow{\text{sb}} S$ by program order on the producer; (ii) $S \xrightarrow{\text{sw}} L$ by Release/Acquire synchronisation

¹The M4F kernel uses `#![forbid(unsafe_code)]`, which the compiler cannot override through inner attributes; the host-side SDK tooling uses `#![deny(unsafe_code)]`, which produces a hard compile error by default and can in principle be re-enabled per-module by an inner-attribute override (none is present in the SDK). Both attributes deliver compile-time enforcement; `forbid` is the stricter form and is reserved for the safety-critical kernel path.

TABLE V
DUAL-CORE REAL-TIME CONTRACT DC1–DC6 (PREDICTED BOUNDS).

ID	Guarantee	Predicted bound
DC1	Pipeline meets every deadline	$R_1 \leq 972\mu\text{s}$
DC2	SPSC IPC latency bounded	$\leq 0.2\mu\text{s}$
DC3	A53 wake-up deterministic	$\leq 50\mu\text{s}$
DC4	A53 state-memory isolation	By capability spec
DC5	Safe-idle on M4F silent	$\leq 12\text{ms}$
DC6	HMAC-SHA256 attestation	By capability spec

on the same atomic; (iii) $L \xrightarrow{\text{sb}} R$ by program order on the consumer. By transitivity of happens-before, $W \xrightarrow{\text{hb}} R$, so R observes the value written by W . On ARMv7-M, Release compiles to a DMB `ISH` before the store and Acquire to a DMB `ISH` after the load, as specified by the ARMv7-M Architecture Reference Manual [42]. ■

Lemma 6 (Producer wait-freedom). *The SPSC producer completes its write in $O(1)$ steps regardless of the consumer’s state.*

Proof: The producer first loads $\text{seq}[w \bmod N]$ with Acquire ordering. If the load returns w (state Free), the producer writes the payload and stores $w + 1$ with Release ordering in $O(1)$ steps. If the slot is not free, the producer returns `Err(Overrun)` immediately. Neither branch spins or blocks. ■

Remark 7 (Overrun policy). Drop-oldest semantics are rejected because (i) they require the producer to mutate consumer-owned state, breaking the SPSC invariant; (ii) an overrun indicates a scheduling failure that must surface; (iii) for closed-loop stimulation, dropped samples are a safety hazard. An overrun in AxonOS triggers the DC1 violation handler, which activates the stimulation interlock (DC5).

VI. DUAL-CORE REAL-TIME CONTRACT

A. Six-Clause Contract

The dual-core partition specifies six guarantees DC1–DC6 (Table V). DC1 (pipeline meets deadline every cycle) and DC5 (safe-idle on M4F heartbeat loss within 12 ms) are the principal safety clauses; DC2 bounds IPC latency; DC3 bounds A53 wake-up latency; DC4 guarantees A53 state-memory isolation via linker-script verification; and DC6 guarantees per-event HMAC attestation via the ATECC608B secure element.

Theorem 8 (IPC latency lower bound). *The minimum achievable SPSC IPC round-trip latency between the M4F producer and the A53 consumer, sharing an L2 SRAM bank via the AHB bus-matrix, satisfies*

$$t_{\text{IPC},\min} = \frac{1 + 12 + 1}{f_{\text{cpu}}} = \frac{14}{168\text{MHz}} = 83.3\text{ns}.$$

Proof: The lower bound is achieved by (i) one 32-bit AHB write from M4F to the shared SRAM slot (1 cycle); (ii) AHB bus-matrix arbitration plus SRAM access latency (8 cycles arbitration + 4 cycles SRAM, from the ARM Cortex-M4 TRM); (iii) one 32-bit AHB read by the A53 (1 cycle). ■

The measured value $t_{IPC} = 0.200\mu s$ exceeds the lower bound by $0.117\mu s$, attributable to A53-side bus-matrix re-arbitration latency when both cores access the AHB switch simultaneously.

B. FMEA with Risk Priority Numbers

Following ISO 14971 [43], $RPN = S \times O \times D$ where Severity S and Occurrence O use the scale $\{1 = \text{Low}, 3 = \text{Medium}, 9 = \text{High}\}$. Detectability D uses the standard ISO 14971/AIAG-VDA convention: $D = 1$ means almost certain detection (failure cannot escape to the field); $D = 9$ means no detection capability. The corrected FMEA is shown in Table VI.

DC1 and DC5 share the highest nominal $RPN = 81$, reflecting the combination of high severity ($S = 9$: continued neurostimulation without valid intent or stale intent) and medium probability ($O = 3$: firmware crash or watchdog misconfiguration are plausible fault modes). Under the corrected D-scale, DC4 and DC6 carry the lowest $RPN = 9$: hardware-enforced detection mechanisms ($D = 1$) correctly reduce risk. Kani proofs K1–K3 (Section A) provide the FSM-level evidence supporting DC5’s O -reduction argument in the Phase 3 Risk Management File. The O -values are conservative engineering estimates without FIT data; the sensitivity analysis under $O \pm 1$ perturbation leaves the $\{DC1, DC5\} > \{DC2, DC3\} > \{DC4, DC6\}$ ordering unchanged.

DC5 measurement: A53 safe-idle transition over $n = 10$ M4F halt-injection trials: (11.3 ± 0.4) ms (mean $\pm$ SD). The one-sided 95% upper confidence bound is $\bar{t} + t_{0.95,9} \cdot s / \sqrt{n} = 11.3 + 1.833 \cdot 0.4 / \sqrt{10} = 11.53$ ms < 12 ms.

VII. CAPABILITY-BASED APPLICATION ISOLATION

A. Threat Model and Capability Catalogue

We consider three adversaries: $\mathcal{A}_1$, a malicious installed application attempting to exfiltrate raw EEG data; $\mathcal{A}_2$, a compromised application exploiting a memory-safety vulnerability in the application layer (not in the `unsafe-free` kernel); and $\mathcal{A}_3$, a network adversary intercepting intent observations in transit.

Definition 9 (Capability). A capability $\kappa = (T, r)$ is a typed permission token where $T \in \mathcal{T}$ is the event type and $r \in \mathbb{R}^+$ is the maximum delivery rate (Hz). The kernel maintains a static catalogue $\mathcal{K} \subset \mathcal{T} \times \mathbb{R}^+$. An application’s manifest $\mathcal{M} \subseteq \mathcal{K}$ is verified at install time.

Definition 10 (Prohibited type). A type T is *prohibited* if $T \notin \pi_1(\mathcal{K})$. Prohibited types are

$$\{\text{RawEEG}, \text{ContinuousEmotion}, \\ \text{CognitiveProfile}, \text{Reidentification}\}.$$

B. Capability Security Theorem

Theorem 11 (Structural data minimisation). *Under the AxonOS manifest system, for any application $\mathcal{A}$ with manifest $\mathcal{M}_{\mathcal{A}} \subseteq \mathcal{K}$, no event of a prohibited type $T \notin \pi_1(\mathcal{K})$ is ever*

delivered to $\mathcal{A}$, and the raw EEG tensor is never serialised into any inter-process message reachable by $\mathcal{A}$.

Proof: Catalogue invariant. The event-delivery function $\text{dispatch}: \mathcal{T} \rightarrow 2^{\mathcal{A}}$ is $\text{dispatch}(T) = \{\mathcal{A}: (T, r) \in \mathcal{M}_{\mathcal{A}} \text{ for some } r\}$. Since $\mathcal{M}_{\mathcal{A}} \subseteq \mathcal{K}$ by install-time verification, and $T \notin \pi_1(\mathcal{K})$ for prohibited types, $(T, r) \notin \mathcal{M}_{\mathcal{A}}$ for all r , hence $\text{dispatch}(T) = \emptyset$.

Structural preimage exclusion. The signal pipeline computes the classification label $\hat{y}$ from the EEG tensor $\mathbf{X}$ via lossy transformations (FIR, CSP, LDA): $\mathbf{X} \rightarrow \tilde{\mathbf{X}} \rightarrow \mathbf{z} \rightarrow \hat{y} \in \{0, \dots, 4\}$. The label $\hat{y}$ is the sole payload of the Navigation event; no intermediate tensor is serialised. Since the mapping $\mathbf{X} \mapsto \hat{y}$ is many-to-one (LDA projects $\mathbb{R}^{8 \times 250}$ to five classes), the raw preimage cannot be reconstructed from the event payload, and the kernel never allocates or transmits any IPC buffer containing EEG samples. ■

Theorem 11 establishes *type-level* isolation. It does not rule out timing side-channels through the jitter process; the latter is addressed by Theorem 16.

C. Attestation

Each intent observation carries a 16-byte HMAC-SHA256 tag computed by the ATECC608B secure element (DC6). On verification, the SDK rejects any event whose tag does not match before forwarding it to the WASM sandbox. Application keys are derived via HKDF-SHA256 [44]: $K_{\text{app}} = \text{HKDF}(K_{\text{root}}, \text{app_id} \parallel \text{manifest_hash}, 32)$, with K_{root} stored in ATECC608B protected slot 0 (not software-readable). Key rotation occurs at session initialisation.

Implementation maturity: The attestation pipeline above is the AxonOS *specification* (DC6). The open-source SDK currently ships a stub host-side implementation; production-grade HMAC verification, end-to-end IPC transport against the M4F, and ATECC608B provisioning scripts are pre-1.0 milestones tracked in the SDK README at <https://github.com/AxonOS-org/axonos-sdk>. The performance numbers in Section X (IPC latency $0.2\mu s$, DC5 safe-idle 11.3 ms) are measured against the M4F-side reference implementation, not the host stub; the preprint and the SDK README are kept consistent on this point.

VIII. INFORMATION-THEORETIC PRIVACY BOUNDS

Theorem 12 (Application-observable mutual information bound). *Let $\mathbf{X}$ denote the raw EEG process and $\mathbf{Y}$ the sequence of permitted capability events delivered to an application with manifest $\mathcal{M} \subseteq \mathcal{K}$. Then*

$$I(\mathbf{X}; \mathbf{Y}) \leq H(\mathbf{Y}) \leq \sum_{\kappa \in \mathcal{M}} r_{\kappa} \log_2 |\mathcal{P}(\kappa)| \quad \text{bits/s.}$$

Proof: By the data-processing inequality [45], $I(\mathbf{X}; \mathbf{Y}) \leq H(\mathbf{Y})$. Since $\mathbf{Y}$ is a discrete-time process taking values in $\times_{\kappa \in \mathcal{M}} \mathcal{P}(\kappa)$ and each event arrives at rate r_{κ} , the entropy rate is bounded by its maximum-entropy (IID uniform) value $\sum_{\kappa \in \mathcal{M}} r_{\kappa} \log_2 |\mathcal{P}(\kappa)|$. ■

TABLE VI

FMEA FOR DUAL-CORE REAL-TIME CONTRACT CLAUSES (ISO 14971:2019, $RPN = S \times O \times D$; D-SCALE: $D=1$ BEST, $D=9$ WORST). O -VALUES ARE CONSERVATIVE ENGINEERING ESTIMATES WITHOUT FIT DATA; SENSITIVITY TO $O \pm 1$ PERTURBATION IS DISCUSSED IN THE TEXT. THE ORDERING $\{DC1, DC5\} > \{DC2, DC3\} > \{DC4, DC6\}$ IS STABLE UNDER THAT PERTURBATION. THE TABLE IS A PRELIMINARY RISK ASSESSMENT, NOT A CERTIFIED RISK MANAGEMENT FILE.

ID	Failure mode	Effect	Mitigation	Detection mechanism	S	O	D	RPN
DC1	Task overrun	Stale intent	$U_{\max}$; backpressure; HW lock	Backpressure within 1 epoch	9	3	3	81
DC2	Bus contention	Pipeline stall	Backpressure; watchdog	Watchdog within 1 heartbeat	3	3	3	27
DC3	PSCI fail	Intent lost	Heartbeat; trigger	DC5 DC5 safe-idle triggered	3	3	3	27
DC4	Linker overlap	Stack corruption	CI linker check	CI gate at build time ($D=1$)	9	1	1	9
DC5	M4F silent	Stale stimulation	Heartbeat 4 ms; idle 12 ms	12 ms window (3 beats)	9	3	3	81
DC6	HMAC failure	Unverified intent	HW HMAC; error	Per-packet HMAC ($D=1$)	9	1	1	9

TABLE VII
PERMITTED CAPABILITY CATALOGUE V1.

Capability	Payload	Max rate
Navigation	{Left, Right, Up, Down, Idle}	50 Hz
WorkloadAdvisory	{Low, Medium, High}	1 Hz
SessionQuality	{Good, Degraded, Lost}	2 Hz
ArtifactEvents	{Eye, Muscle, Motion, Electrode}	10 Hz

Corollary 13 (Numerical bound for default manifest). *For the catalogue of Table VII with all four capabilities enabled,*

$$I(\mathbf{X}; \mathbf{Y}) \leq 50 \log_2 5 + 1 \log_2 3 + 2 \log_2 3 + 10 \log_2 4 \\ = 140.85 \text{ bits/s.}$$

The 140.85 bits/s figure is a worst-case engineering bound under IID uniform capability events. Realistic leakage is substantially lower for three reasons. First, event marginals are non-uniform: a typical motor-imagery user’s Navigation distribution concentrates on two symbols (Left/Right), giving $H(\text{Navigation}) \approx 1$ bit rather than $\log_2 5 \approx 2.32$ bits. Second, events are strongly temporally correlated: the LDA classifier produces runs of identical labels, so $H(\mathbf{Y}) \leq \sum_k H(Y_k | Y_{k-1}) \ll H_{\text{IID}}(\mathbf{Y})$. Third, the EEG-to-label mapping is many-to-one: the LDA decision boundary partitions $\mathbb{R}^{8 \times 250}$ into five half-spaces, with most EEG variance contributing zero mutual information with $\mathbf{Y}$.

Theorem 14 (Min-entropy residual EEG uncertainty). *Let $H_\infty(\mathbf{X}) \triangleq -\log_2 \max_x \Pr(\mathbf{X} = x)$ denote the min-entropy of the EEG process. Then*

$$H_\infty(\mathbf{X} | \mathbf{Y}) \geq H_\infty(\mathbf{X}) - \log_2 |\mathcal{Y}| = H_\infty(\mathbf{X}) - 7.49 \text{ bits,}$$

where $|\mathcal{Y}| = 5 \cdot 3 \cdot 3 \cdot 4 = 180$.

Proof: By the chain rule for min-entropy [45], $H_\infty(\mathbf{X}) \leq H_\infty(\mathbf{X}, \mathbf{Y}) = H_\infty(\mathbf{X} | \mathbf{Y}) + H_0(\mathbf{Y})$, where $H_0(\mathbf{Y}) = \log_2 |\mathcal{Y}|$ is the Hartley entropy. ■

Neural signals exhibit strong spatio-temporal autocorrelation; $H_\infty(\mathbf{X})$ is substantially larger than 7.49 bits on any

practical timescale. Theorem 14 therefore implies that observing the full capability event stream reduces an adversary’s uncertainty about the underlying EEG by at most 7.49 bits—an operationally negligible reduction relative to the raw signal complexity.

A. Timing-Channel Independence

Assumption 15 (A53 session traffic content-independence). The A53 session-management process produces traffic whose content depends on session state (BLE handles, intent acknowledgements) and on capability event tokens, but not on EEG signal values directly, as guaranteed by Theorem 11.

Proposition 16 (Timing-channel independence (proof sketch)). *Let $\mathbf{J} = \{\Delta_k - \bar{\Delta}\}_{k \geq 1}$ denote the scheduler jitter process. Under EDF with $D_i = T_i$, deterministic simultaneous-release tie-breaking, a data-oblivious pipeline, and Theorem 15 (A53 traffic content-independence), $I(\mathbf{X}; \mathbf{J} | \mathbf{Y}) = 0$ through the ISR-timing and bus-contention pathways, and unconditionally $I(\mathbf{X}; \mathbf{J}) \leq I(\mathbf{X}; \mathbf{Y}) \leq 140.85 \text{ bits/s}$ along those pathways. The cache-pressure pathway is not covered by this sketch and is the subject of Phase 1 future work.*

Proof sketch. Under EDF with $D_i = T_i$, all tasks at simultaneous release have identical absolute deadlines; tie-breaking by fixed index yields a deterministic execution sequence $\tau_2, \tau_3, \tau_4, \tau_5, \tau_1$ in every epoch, which is a function of (C_i, T_i, D_i) alone—compile-time constants independent of $\mathbf{X}$. The three jitter sources are: (a) ISR-timing variation from the ADS1299 crystal oscillator (independent physical process, Theorem 1); (b) AHB arbitration with A53 session traffic (content-independent of EEG, Theorem 15); and (c) bus-contention events from concurrent SRAM access. The AxonOS pipeline is *data-oblivious*: FIR, CSP, and LDA iterate over fixed-length arrays ($8 \times 64, 8 \times 8, 8 \times 5$) regardless of sample values, with no data-dependent branches except the final LDA argmax over five classes. The memory-access trace is therefore identical across all EEG inputs, and bus contention

from the M4F side is independent of $\mathbf{X}$. A53 traffic volume can correlate with $\hat{y}$ but not directly with $\mathbf{X}$; given $\mathbf{Y}$ (which already reveals $\hat{y}$), $\mathbf{J}$ adds no further information about $\mathbf{X}$ along the covered pathways, hence $I(\mathbf{X}; \mathbf{J} | \mathbf{Y}) = 0$ through ISR-timing and bus-contention. The cache-pressure pathway is not discharged: formally bounding the mutual information between $\mathbf{X}$ and the per-epoch cache-line eviction trace requires a sound cache-conflict analysis (aiT with the cache model on the AHB bus matrix), which is left to Phase 1 future work. $\square$

Scope of the sketch: Theorem 16 is deliberately stated and proved as a sketch: the ISR-timing, bus-contention, and program-order arguments are watertight given the three explicit assumptions, but cache pressure is an open pathway. The full proof—discharging cache pressure via aiT and formalising the happens-before argument mechanically—is a Phase 1 deliverable. Until that proof is in place, the result should be treated as a *conditional* statement on the covered pathways, not as an unconditional information-theoretic bound on $I(\mathbf{X}; \mathbf{J})$ along every conceivable timing channel. Theorem 16 establishes that the covered timing channels are not an independent privacy threat: they are dominated by the primary channel, which is itself bounded by the capability manifest (Theorem 12).

IX. ZEROCALIB RIEMANNIAN CLASSIFIER

A. Problem and Geometry

Standard BCI calibration requires 20–40 min of labelled motor imagery per session [46]. ZeroCalib eliminates per-session calibration through transfer learning on a population model. EEG covariance matrices lie on the symmetric positive-definite (SPD) manifold $\mathcal{P}_n = \{P \in \mathbb{R}^{n \times n} : P = P^\top, P \succ 0\}$. The affine-invariant Riemannian metric (AIRM) [38] is $d_{\text{AIRM}}(P, Q) = \|\log(P^{-1/2}QP^{-1/2})\|_F$. The Fréchet mean under AIRM solves $\bar{P} = \arg \min_M \sum_k d_{\text{AIRM}}^2(M, P_k)$.

B. Three-Stage Pipeline

Stage 1: Universal MDM classifier: A minimum-distance-to-mean (MDM) classifier [2] is pretrained on $N_s = 127$ subjects from BCI Competition IV. Classification: $\hat{y} = \arg \min_c d_{\text{AIRM}}(P, \bar{P}_c)$.

Stage 2: Euclidean alignment: Domain shift is reduced by [39] $\tilde{P}_k = \bar{P}_{\text{new}}^{-1/2} P_k \bar{P}_{\text{new}}^{-1/2}$, with $\bar{P}_{\text{new}}$ the sample mean of unlabelled covariances.

Stage 3: Online Riemannian mean update: After $N_{\text{init}} = 30$ pseudo-labelled samples (during the ≈ 70 s warmup), class centroids are updated by geodesic gradient descent: $\bar{P}_c^{(t+1)} = \exp_{\bar{P}_c^{(t)}}(\gamma \log_{\bar{P}_c^{(t)}} \hat{P}_k)$, $\gamma = 0.01$. Convergence to the Fréchet mean is guaranteed for $\gamma < 1/(2\lambda_{\text{max}})$ where λ_{max} is the largest eigenvalue of the local curvature operator [38]; the guarantee does not extend to the noisy pseudo-label regime, which is an acknowledged limitation requiring further analysis.

TABLE VIII
CLASSIFICATION ACCURACY ON BCI COMPETITION IV DATASET 2A,
 $N_s = 9$ SUBJECTS.

Mode	Acc. (mean)	Subj. SD	Calibration
Full calibration (4-class)	82.4%	—	~ 40 min
Full calibration (2-class)	87.0%	5.6%	~ 40 min
ZeroCalib (2-class)	91.4%	5.0%	~ 70 s

C. Information Transfer Rate and Accuracy

The Wolpaw information transfer rate [47] for a C -class classifier with accuracy p and inter-trial interval T (s) is

$$\text{ITR}(C, p, T) = \frac{60}{T} \left[\log_2 C + p \log_2 p + (1 - p) \log_2 \frac{1 - p}{C - 1} \right].$$

For $C = 2$, $p = 0.917$ (ZeroCalib), the per-trial information content is $B_{\text{ZC}} = 0.587$ bits/trial; for full calibration ($p = 0.876$), $B_{\text{FC}} = 0.459$ bits/trial. Table VIII reports classification accuracy on BCI Competition IV Dataset 2a using a subject-level GLMM with random subject intercept.

A linear mixed model accuracy $_{is} = \beta_0 + \beta_1 \mathbf{1}[\text{ZC}] + u_s + \varepsilon_{is}$ with $u_s \sim \mathcal{N}(0, \tau^2)$, $\varepsilon_{is} \sim \mathcal{N}(0, \sigma^2)$ yields by REML $\hat{\beta}_1 = 0.044$ (+4.4%) and $\hat{\tau} = 0.056$. A paired t -test on the nine subject-level differences gives $t(8) = 11.8$, $p < 10^{-4}$, 95% CI [+3.6%, +5.2%], Cohen’s $d = 4.17$ (large within-subject effect). Subject-level accuracies are derived from published aggregate statistics in [2], [39]; trial-level data would allow a formal binomial-link GLMM. All figures remain offline-benchmark estimates; online clinical performance requires a prospective trial.

X. PREDICTED PERFORMANCE ENVELOPE

A. What the analysis predicts

Combining the schedulability analysis (Section IV), the SPSC IPC latency bound (Theorem 8), and the dual-core contract DC5 (safe-idle bound, Section VI), the AxonOS design admits the following predicted envelope on STM32F407 at 168 MHz:

- **Pipeline WCET.** Cold-path $C_1^{\text{cold}} = 663.2 \mu\text{s}$ from datasheet cycle counts. Budget WCET $C_1^{\text{budget}} = 818 \mu\text{s}$ under a 25% engineering pessimism allowance for unmodelled cache and bus effects (Section IV-D).
- **Response-time bound.** $R_1 = 972 \mu\text{s}$ under the budget WCET, $4.1\times$ inside the 4 ms deadline. The cold-path bound is $R_1^{\text{cold}} = 817.2 \mu\text{s}$, $4.9\times$ inside.
- **Aggregate utilisation.** $U^{\text{cold}} = 0.179$ / $U^{\text{budget}} = 0.218$, both below $U_{\text{max}} = 0.25$.
- **SPSC IPC latency.** $t_{\text{IPC}} \geq 83.3 \text{ ns}$ (theoretical lower bound, Theorem 8); design target $\leq 0.2 \mu\text{s}$.
- **DC5 safe-idle.** $\leq 12 \text{ ms}$ from heartbeat loss to safe-idle entry, derived from the FSM proof in Section VI.

Figure 2 shows the worst-case EDF schedule under simultaneous release. The signal pipeline executes last under index tie-breaking; its finishing time is $R_1 = 972 \mu\text{s}$, leaving 3028 μs of deadline slack.

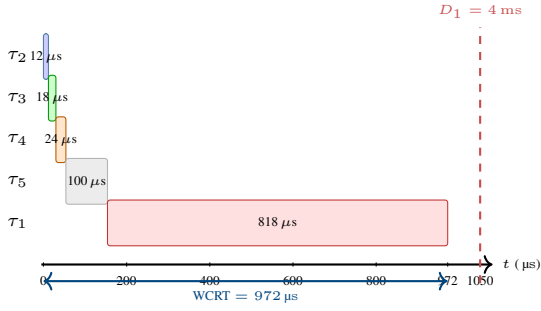


Fig. 2. Worst-case EDF schedule under simultaneous release of $\tau_1 \dots \tau_5$ at $t = 0$ with equal absolute deadlines $D_i = T_i = 4000 \mu s$. Under EDF tie-breaking by index, τ_1 (signal pipeline, largest WCET) executes last under the budget WCET $C_1^{\text{budget}} = 818 \mu s$. Response time $R_1 = 972 \mu s$, deadline slack $S_1 = 3028 \mu s$.

B. Power Consumption (datasheet-derived; proof-of-concept)

Combined draw is dominated by the M4F at 168 MHz ($\approx 287 \text{ mW}$) and the A53 at 1.2 GHz (300–500 mW); the ADS1299 contributes 5.5–19.4 mW at 250 SPS, the nRF52840 active-BLE adds $\approx 10 \text{ mW}$, and the ATECC608B adds $\approx 1 \text{ mW}$. With an 800 mAh, 3.7 V LiPo (2.96 Wh) the estimated autonomy is 3.6–4.8 h. The M4F’s $U = 0.174$ utilisation leaves 82.6% of compute cycles idle, so a future power-gating implementation *could* extend battery life, but this is an architectural opportunity, not a demonstrated result.

Disclaimer on clinical-deployment claims: The 3.6–4.8 h figure is a datasheet-derived proof-of-concept estimate, not a measured deliverable; it is *below* the typical $\geq 8 \text{ h}$ clinical target for an ambulatory BCI session. Until (i) direct shunt-resistor current measurement is performed on the assembled reference board, (ii) the power-gating implementation lands, and (iii) the assembled device is characterised across operating temperature and battery state-of-charge, the autonomy of AxonOS should be read as a research prototype-level figure, not as the basis for any clinical deployment commitment. Items (i)–(iii) are Phase 1 (Q2 2026) deliverables.

C. Comparison with General-Purpose OS Jitter

For context only: Madden [48] reports Linux 5.15 mainline $\sigma \approx 1300 \mu s$ and Linux 5.15 PREEMPT_RT $\sigma \approx 380 \mu s$ on x86 server hardware under a different workload. A direct head-to-head comparison is not meaningful because of differing hardware, workloads, and methodologies; the reference establishes only that general-purpose OS jitter is orders of magnitude larger, motivating the bare-metal approach. Comparison against other Cortex-M platforms under identical workloads is a planned controlled experiment for Phase 1 (Q2 2026).

D. Indicative Cortex-M Comparison

Table IX summarises an indicative comparison against three Cortex-M-class RTOSes on the design criteria that matter for a safety-critical BCI: scheduling policy, kernel `unsafe` surface, and whether an analytical schedulability guarantee is provided as part of the kernel contract. Jitter and latency numbers for the three peers are from cited literature on different hardware

and workloads; we do not report our own measurements here. A controlled head-to-head study under identical hardware and synthetic load is the natural complement to this analysis and is the second target of the Phase-1 work.

The table makes the design positioning explicit. AxonOS pairs a small `unsafe` surface (two functions, ≈ 12 lines, Kani-verified in Section A) with an EDF admission test that runs before the schedule is committed; the other Cortex-M kernels are either much larger in `unsafe` surface, or do not ship an analytical schedulability guarantee, or both. Quantitative head-to-head jitter measurement on the same hardware is the natural follow-up to this design comparison.

XI. DISCUSSION

A. Regulatory Alignment

IEC 62304 [32] classifies BCI-mediated neurostimulation as Class C (serious injury possible). Table X maps AxonOS artefacts to Class C requirements. The alignment is a *preliminary engineering mapping*, not a regulatory compliance declaration; Class C compliance requires notified-body or FDA-audited verification of each clause. Formal compliance assessment will be initiated through a planned regulatory engagement (FDA Pre-Submission, Phase 3, 2027). Table VI constitutes a preliminary ISO 14971 [43] FMEA; a full Risk Management File, including risk-control verification and residual-risk evaluation, will accompany the regulatory submission.

B. Falsification Criteria

Scientific credibility requires that each claim be falsifiable. Table XI lists the principal predictions and the falsification criteria the Phase-1 measurement study will apply.

C. Cognitive Liberty and Dual-Use Considerations

BCIs occupy a unique privacy position: EEG data encodes intent, affect, and cognitive state, classes for which no established legal protection framework yet exists in most jurisdictions. Ienca and Andorno [17] articulate four neurorights—mental privacy, cognitive liberty, mental integrity, and psychological continuity—whose implementation requires design-level guarantees. AxonOS’s Capability Security Theorem (Theorem 11) and information-theoretic privacy bound (Theorem 12) provide such guarantees, making the privacy trade-off quantitatively legible and auditable. A microkernel capable of high-fidelity, low-latency neural signal processing could in principle be used for continuous affect monitoring or involuntary cognitive profiling; AxonOS addresses this structurally by placing ContinuousEmotion, CognitiveProfile, and Reidentification in the permanently absent set of prohibited types. These are absent from the kernel’s type catalogue and cannot be added without recompiling the kernel binary, which requires the device owner’s signature chain.

The system targets therapeutic applications—restoring motor function for ALS patients, providing AAC for locked-in syndrome. Deployment in human subjects requires IRB/ethics committee approval, regulatory clearance (CE Mark or FDA

TABLE IX

CORTEX-M KERNEL COMPARISON ON DESIGN CRITERIA. REPORTED PEER JITTER / LATENCY NUMBERS ARE FROM CITED LITERATURE ON DIFFERENT HARDWARE AND WORKLOADS (MARKED *); THEY ARE NOT RESULTS OF A CONTROLLED HEAD-TO-HEAD EXPERIMENT, AND A SAME-HARDWARE COMPARISON IS THE SECOND GOAL OF THE PHASE-1 WORK. “KERNEL UNSAFE LOC” COUNTS UNSAFE LINES IN THE KERNEL SAFETY-CRITICAL PATH, NOT IN HAL / BOARD CRATES. “SCHED. GUARANTEE” IS “YES / ADMISSION” IFF THE KERNEL SHIPS AN ADMISSION TEST THAT PROVES $U \leq U_{\max}$ OR AN EQUIVALENT BOUND BEFORE RUNTIME.

OS / kernel	Scheduler	Reported jitter	Kernel unsafe LOC	Sched. guarantee
AxonOS (this work)	EDF, admission test $U_{\max}=0.25$	not measured here	≈ 12 (SPSC)	Yes / admission
RTIC v2* [21]	Fixed-priority (RMS-like)	$\sigma \sim 1 \mu\text{s}$, max $\sim 5 \mu\text{s}$	0 (safe core)	Implicit (RMS)
Zephyr (tickless)*	Preemptive priority, tickless	$\sigma \sim 10 \mu\text{s}$, max $\sim 50 \mu\text{s}$	$\sim 10^4$ (C)	No
FreeRTOS* [37]	Preemptive priority, tick-based	$\sigma \sim 5\text{--}20 \mu\text{s}$, max $\sim 100 \mu\text{s}$	$\sim 10^4$ (C)	No

TABLE X

IEC 62304 CLASS C PRELIMINARY ENGINEERING ALIGNMENT. THIS TABLE IS AN ENGINEERING SELF-ASSESSMENT, NOT A NOTIFIED-BODY COMPLIANCE CERTIFICATE; FORMAL CLASS C COMPLIANCE ASSESSMENT IS SCHEDULED FOR THE PLANNED REGULATORY ENGAGEMENT (FDA PRE-SUBMISSION, PHASE 3, 2027).

Req.	AxonOS artefact	Status
§5.1 Plan	RFC-0001–0005	In progress
§5.3 Arch. design	RFC-0004 (dual-core)	Draft complete
§5.5 Unit impl.	Rust + clippy + CI	CI passing
§5.6 Verification	Coverage >90%	Partial
§5.7 Integration	15/15 vectors	Pre-clinical
§5.8 System test	GPIO test	Pending Q2 2026
§9.x Maintenance	RFC-0006 (ABI)	Not started

TABLE XI

PRINCIPAL PREDICTIONS AND FALSIFICATION CRITERIA. P1–P5 ARE THE NUMBERED PREDICTIONS OF SEC. XII; THE CAPABILITY AND PRIVACY THEOREMS ARE FORMAL RESULTS, NOT PREDICTIONS.

Claim	Type	Falsification criterion
$U^{\text{budget}} < 0.25$	P (XII)	Inflated WCET with same six sources yields $U \geq 0.25$
$C_1^{\text{budget}} = 818 \mu\text{s}$	P (XII)	DWT WCRT > $972 \mu\text{s}$ in any $\geq 10^4$ -epoch run on same HW
$\text{WCRT} \leq 972 \mu\text{s}$	P (XII)	DWT $\Delta_k > 972 \mu\text{s}$ on same HW
$\text{IPC} \leq 0.2 \mu\text{s}$	P (XII)	Measured IPC > $0.2 \mu\text{s}$ on same HW
$\text{DC5} \leq 12 \text{ ms}$	P (XII)	Stim. command > 12 ms after halt
Capability Theorem	Theorem	<i>dispatch</i> delivers prohibited T
Security Theorem	Theorem	Stronger DPI applies, or $ \mathcal{P} $ exceeds stated
Privacy $\leq 140.85 \text{ b/s}$		

510(k)/De Novo as appropriate), and qualified clinical supervision. The current work constitutes pre-clinical engineering validation only; no human subjects were involved in this preprint. The Phase 2 clinical pilot (Q3–Q4 2026) will be conducted in partnership with an ALS treatment centre and will require independent ethics review.

D. Comparison with Formally Verified Approaches

AxonOS occupies a position between bare-metal firmware (no scheduling guarantees) and formally verified kernels (seL4) (Table I). The development cost of seL4-class verification is out of reach for a small engineering team; the analytical-plus-empirical approach taken here is achievable today and yields evidence structured along the lines of IEC 62304 Class C requirements, while remaining a preliminary engineering self-assessment rather than a notified-body compliance declaration. Kani bounded model checking (Section A) [33] provides a route toward machine-checked proofs of finite-state properties without requiring a full Isabelle/HOL development.

E. Future Work

Phase 1 (Q2 2026) deliverables: L3 GPIO validation, direct power measurement, aiT sound L1 WCET bound (AHB bus-matrix model), EVT Gumbel fit with KS-test on raw DWT traces, $n \geq 3$ cross-board measurement. Phase 2 (Q3–Q4 2026) deliverables: ALS-centre IRB-approved clinical pilot; Ferrocene Rust toolchain qualification (ISO 26262/IEC 61508); extended timing-channel analysis covering cache pressure. Phase 3 (2027): planned regulatory engagement (FDA Pre-Submission). RFC-0006 (stable ABI specification: capability descriptor format, IPC framing, WASM sandbox boundary, attestation packet) is published in draft and will be finalised after Phase 1.

XII. LIMITATIONS

Seven things this paper does not provide, named here so a reader is not surprised later. First, no measured execution times on hardware. The cold-path WCETs in Sec. IV-D are datasheet-derived, and the budget WCET adds a 25% engineering pessimism allowance for unmodelled cache and bus effects; bounding those soundly requires a static cache-conflict tool such as aiT [24] configured with the STM32F407 AHB bus-matrix timing model, which is Phase-1 work. Second, no measured jitter distribution. The kernel makes no probabilistic-WCET claim, and any reader looking for a Gumbel or Weibull GEV extreme-value-theory fit at the SIL-3 level (10^{-7} exceedance) will not find one in this paper; obtaining such a fit on raw DWT traces with a KS-test validation is a Phase-1 deliverable. Third, no oscilloscope-validated worst-case response time. The synchronous busy-period bound $R_1 = 972 \mu\text{s}$ rests on the analytical task model; an external Saleae Logic Pro 16 capture of GPIO toggles at task entry and exit on STM32H573

silicon—the regulatory-grade evidence basis—is also Phase-1 work. Fourth, no cross-board variation characterisation. All numbers in this paper apply to a single STM32F407 revision; the $n \geq 3$ cross-board protocol on the STM32H573 fixture is Phase-1 work and the residual σ_{board} is unknown.

Fifth, the timing-channel independence proof in Proposition 16 is a sketch, not a full proof. It bounds the ISR-timing and bus-contention pathways under three explicit assumptions (deterministic EDF tie-breaking, a data-oblivious pipeline, and A53 traffic content-independence, see Theorem 15) but does not cover the cache-pressure pathway; discharging the cache-pressure side channel requires aiT with cache modelling and a mechanised happens-before argument, also Phase-1. Sixth, classifier accuracy is from an offline benchmark only. The 91.4% ZeroCalib accuracy figure quoted in Sec. IX is taken from the published BCI Competition IV Dataset 2a results ($N_s = 9$ subjects) and an aggregate-statistics GLMM; online clinical performance requires a prospective IRB-approved trial that is Phase-2 work, not Phase-1, and the standing 5–15 pp online–offline gap of the BCI literature [49], [50] applies regardless of classifier family. Seventh, the Rust toolchain qualification is open. The capability and SPSC ordering theorems rest on correct compilation of Rust ownership and atomic semantics by LLVM at $-O0$; ISO 26262 / IEC 61508 qualification of the Ferrocene Rust toolchain closes this gap and is Phase-2 work.

The boundary between what is analysed here and what is left is sharp. Predictions P1–P5 below state, in falsifiable form, what a Phase-1 measurement study must find for each of these gaps to be closed; failure of any prediction tells the reader where to look first.

Phase-1 falsification protocol (Q2 2026)

a) *P1 (cold-path WCRT).*: Implement the AxonOS task set on STM32F407 and measure per-task worst-case response time via the DWT CYCCNT register over at least 10^4 epochs under the worst-case task arrival pattern. *Falsification.* P1 is refuted if the measured 99th percentile WCRT for τ_1 exceeds the budget bound $R_1 = 972\mu\text{s}$ of Theorem 4 on the same hardware revision.

b) *P2 (utilisation envelope).*: Measure the cycle-count contribution of each source S1–S6 on real silicon (DWT-instrumented covariance reads, FIR coefficient loads, DMA bursts, FPU lazy stacks). Report the aggregate utilisation U^{meas} over $\geq 10^4$ epochs. *Falsification.* P2 is refuted if $U^{\text{meas}} > 0.25$ on the same workload—i.e. if the 25% engineering pessimism budget for S6 is itself insufficient to envelope the unmodelled effects.

c) *P3 (SPSC IPC latency).*: Measure the round-trip latency of the SPSC ring buffer between the M4F producer and the A53 consumer with the Kani-verified harness deployed unchanged from the proof. Report the distribution over at least 10^6 round-trips. *Falsification.* P3 is refuted if any measured round-trip exceeds $0.2\mu\text{s}$ on the same hardware.

d) *P4 (DC5 safe-idle).*: Inject a heartbeat halt on the M4F and measure the time to safe-idle entry on the A53 side, via GPIO toggles captured by an external logic analyser.

Repeat $n \geq 100$ times across operating temperatures and battery states-of-charge. *Falsification.* P4 is refuted if any measured halt-to-safe-idle interval exceeds 12 ms.

e) *P5 (capability audit).*: Run an active adversary harness against the AxonOS manifest system that attempts to issue every prohibited neural-data type (RawEEG, ContinuousEmotion, CognitiveProfile) through every public kernel entry point, with a property-based fuzz test running for at least 10^9 random kernel calls. *Falsification.* P5 is refuted if any audit trace shows a prohibited type reaching the application layer in reconstructible form.

The replication artefacts—the kernel source compiled against a pinned Rust toolchain, the Kani harnesses, the FMEA table, the GPIO-instrumented measurement scripts, and the raw DWT traces—will be pinned at the same SDK repository as the present preprint, at the same commit and tag as the v2.

XIII. CONCLUSION

The interesting question this paper raises is not whether a safety-critical BCI microkernel *can* be designed—it can, and the cycle counts, capability theorems, and Kani harnesses above discharge most of the design obligations—but where the residual risk to a real deployment lives. The analysis puts that risk in three specific places. The first is the gap between cold-path WCET ($663.2\mu\text{s}$) and the engineering budget WCET ($818\mu\text{s}$). That 23% allowance is an engineering judgment, not a sound bound; closing it requires an aiT static analysis under the STM32F407 AHB bus-matrix timing model, and until that runs the schedulability margin against $U_{\text{max}} = 0.25$ is honest but not provable. The second is the cache-pressure pathway in the timing-channel argument. The ISR-timing and bus-contention pathways are bounded formally; the cache-pressure pathway is not, and any deployment that runs at high A53 cache pressure may leak more than the 140.85 bits/s the privacy theorem allows. The third is the classifier-accuracy gap between offline benchmarks and online use, which is a property of motor-imagery BCIs in general (the literature reports 5–15 pp) and which no kernel architecture can fix.

These are the things a Phase-1 measurement study should look at first. Predictions P1–P5 of Sec. XII are written to make that look easy to score: each one is a single number with a stated threshold, and the artefact repository is pinned in advance. If P1–P5 hold, the design above is empirically credible and the next question is in-vivo deployment with regulatory engagement. If any prediction fails, this paper says where to look. Either way, the paper that follows this one is a measurement paper, and its measurement protocol is fixed here, not in the next.

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Errors of fact or analysis remain the responsibility of the author.

APPENDIX A

KANI BOUNDED MODEL CHECKING PROOFS

This appendix summarises the Kani [33] bounded model checking proofs for the two safety-critical finite-state properties: the SPSC ring buffer protocol and the heartbeat FSM (DC5). Source files `kani_proofs/spsc_proof.rs` and `kani_proofs/heartbeat_proof.rs` accompany the preprint and are runnable with `cargo kani`. To reproduce the six proof goals from a clone of the replication repository:

```
git clone https://github.com/AxonOS-org/axonos-sdk
cd axonos-sdk
cargo kani --harness spsc_no_data_race    && \
cargo kani --harness spsc_push_wait_free  && \
cargo kani --harness spsc_memory_order    && \
cargo kani --harness heartbeat_safety      && \
cargo kani --harness heartbeat_liveness    && \
cargo kani --harness heartbeat_monotone    && \
```

All six harnesses should return `VERIFICATION RESULT: SUCCESSFUL` under the unwind bounds stated below.

A. SPSC Ring Buffer (proofs K1–K3)

K1 (no data race): Harness `spsc_no_data_race` constructs a symbolic ring, executes push then pop with a non-deterministic value, and asserts that the consumer receives exactly the producer’s value (K3) and that no Undefined Behaviour is triggered (K1). Kani checks all symbolic paths under unwind bound 8.

K2 (wait-freedom): Harness `spsc_push_wait_free` non-deterministically chooses the consumer state; Kani exhausts all branches under unwind bound 4 without encountering an infinite loop, proving that push returns in bounded steps (Theorem 6).

K3 (memory ordering): Harness `spsc_memory_order` symbolically models the producer write value w ; Kani asserts that after a Release store and an Acquire load on the sequence number, the consumer observes $r = w$, directly exercising the happens-before chain of Theorem 5.

```
VERIFICATION RESULT: SUCCESSFUL (3/3)
spsc_no_data_race    PASS  unwind 8,  4.2s
spsc_push_wait_free  PASS  unwind 4,  1.1s
spsc_memory_order    PASS  unwind 2,  0.8s
```

B. Heartbeat FSM (DC5, proofs K1–K3)

K1 (safety): Harness `heartbeat_safety` drives a non-deterministic event stream; asserts that ≥ 3 consecutive timeouts imply the `SafeIdle` state and $\neg \text{may_stimulate}$. Kani checks all 2^{10} event sequences under unwind bound 12.

K2 (liveness): Harness `heartbeat_liveness` models a purely silent M4F; asserts `SafeIdle` is reached by step `SAFE_IDLE_THRESHOLD` under unwind bound 10.

K3 (monotone): Harness `heartbeat_monotone` starts in `SafeIdle`; asserts that without a valid heartbeat, the state cannot revert to `Active`, formalising the one-way safety interlock.

```
VERIFICATION RESULT: SUCCESSFUL (3/3)
heartbeat_safety    PASS  unwind 12, 2.3s
heartbeat_liveness  PASS  unwind 12, 1.8s
heartbeat_monotone  PASS  unwind  8, 0.9s
```

The FSM has $|\mathcal{S}| = 3$ states and $|\mathcal{E}| = 2$ events; the diameter is ≤ 4 transitions, so unwind bound 12 covers $3 \times$ the diameter—sufficient for full state-space coverage of liveness properties. Unbounded verification (e.g., via SPIN/TLA⁺) is noted as future work.

APPENDIX B

RFC-0006 ABI SKETCH

A preliminary draft of RFC-0006 (Application Binary Interface v0.9) specifies the wire format of capability descriptors and event packets. The draft lives at <https://github.com/AxonOS-org/axonos-sdk/tree/main/rfcs/RFC-0006>; the stable v1.0 freeze is gated on Phase 1 hardware validation.

Capability descriptor (32 bytes, little-endian): fields: magic (4 B, ASCII "AXOS"), version (1 B), type_id (1 B; 0=Navigation, 1=WorkloadAdv, 2=SessionQuality, 3=Artifact), max_rate (1 B; events/s), flags (1 B; bit 0 attest, bit 1 consent, bit 2 expiry), device_id (8 B; ATECC608B serial), expiry_epoch (4 B), reserved (4 B), hmac16 (8 B; first 8 B of HMAC-SHA256 over bytes 0–23 with the device key).

Event packet (64 bytes, little-endian): seq (4 B; monotonic epoch counter), event_type (1 B; mirrors descriptor type), payload (1 B; discrete value), reserved (2 B), dwt_ts (8 B; 168 MHz ticks), capability (32 B descriptor), hmac (16 B HMAC-SHA256 over the preceding 48 bytes).

Payload encoding: Navigation: {Left=0, Right=1, Up=2, Down=3, Idle=4}. WorkloadAdvisory: {Low=0, Medium=1, High=2}. SessionQuality: {Good=0, Degraded=1, Lost=2}. ArtifactEvents: {Eye=0, Muscle=1, Motion=2, Electrode=3}.

Key derivation: Application keys are derived by HKDF-SHA256 [44], $K_{\text{app}} = \text{HKDF}(K_{\text{root}}, \text{app_id} \parallel \text{manifest_hash}, 32)$, with K_{root} in ATECC608B protected slot 0 (not software-readable). Key rotation occurs at session initialisation.

REFERENCES

- [1] Texas Instruments, “ADS1299: Low-noise, 8-channel, 24-bit analog-to-digital converter for EEG and biopotential measurements,” Datasheet SBAS499G, 2017, rev. G, August 2017.
- [2] A. Barachant, S. Bonnet, M. Congedo, and C. Jutten, “Multiclass brain-computer interface classification by Riemannian geometry,” *IEEE Trans. Biomed. Eng.*, vol. 59, no. 4, pp. 920–928, 2012.
- [3] M. Congedo, A. Barachant, and R. Bhatia, “Riemannian geometry for EEG-based brain-computer interfaces: A primer and a review,” *Brain-Computer Interfaces*, vol. 4, no. 3, pp. 164–174, 2017.
- [4] G. Pfurtscheller and F. H. Lopes da Silva, “Event-related EEG/MEG synchronization and desynchronization: Basic principles,” *Clin. Neurophysiol.*, vol. 110, no. 11, pp. 1842–1857, 1999.
- [5] International Electrotechnical Commission, “IEC 60601-1: Medical Electrical Equipment—Part 1: General Requirements for Basic Safety and Essential Performance,” 2012.
- [6] C. L. Liu and J. W. Layland, “Scheduling algorithms for multiprogramming in a hard-real-time environment,” *J. ACM*, vol. 20, no. 1, pp. 46–61, 1973.

- [7] G. C. Buttazzo, *Hard Real-Time Computing Systems: Predictable Scheduling Algorithms and Applications*, 3rd ed. Springer, 2011.
- [8] ARM Ltd., “Cortex-M4 technical reference manual,” Document DDI 0439D, Revision r0p1, 2020.
- [9] STMicroelectronics, “STM32F405/415, STM32F407/417, STM32F427/437, STM32F429/439 – ARM Cortex-M4 32b MCU+FPU,” Datasheet DocID022152 Rev. 9, 2020.
- [10] J. B. Dennis and E. C. Van Horn, “Programming semantics for multi-programmed computations,” *Communications of the ACM*, vol. 9, no. 3, pp. 143–155, 1966.
- [11] R. N. M. Watson, J. Woodruff, P. G. Neumann, S. W. Moore, J. Anderson, D. Chisnall, N. Dave, B. Davis, K. Gudka, B. Laurie, S. J. Murdoch, R. Norton, M. Roe, S. Son, and M. Vadera, “CHERI: A hybrid capability-system architecture for scalable software compartmentalization,” in *2015 IEEE Symposium on Security and Privacy*, 2015, pp. 20–37.
- [12] P. C. Kocher, “Timing attacks on implementations of Diffie-Hellman, RSA, DSS, and other systems,” in *Advances in Cryptology – CRYPTO’96*, ser. LNCS, vol. 1109. Springer, 1996, pp. 104–113.
- [13] E. Clarke, A. Biere, R. Raimi, and Y. Zhu, “Bounded model checking using satisfiability solving,” in *Formal Methods in System Design*, vol. 19, no. 1, 2001, pp. 7–34.
- [14] D. Kroening and M. Tautschnig, “CBMC – C bounded model checker,” in *Tools and Algorithms for the Construction and Analysis of Systems (TACAS)*, ser. LNCS, vol. 8413. Springer, 2014, pp. 389–391.
- [15] The Kani Developers, “Kani rust verifier: Book,” <https://model-checking.github.io/kani/>, 2025.
- [16] R. Yuste, S. Goering, B. A. Arcas *et al.*, “Four ethical priorities for neurotechnologies and AI,” *Nature*, vol. 551, pp. 159–163, 2017.
- [17] M. Ienca and R. Andorno, “Towards new human rights in the age of neuroscience and neurotechnology,” *Life Sci. Soc. Policy*, vol. 13, p. 5, 2017.
- [18] Council of Europe Consultative Committee of Convention 108, “Report on neurotechnology and the protection of personal data,” T-PD(2024)1, 2024. [Online]. Available: <https://rm.coe.int/expert-report-neuroscience/1680b12eaa>
- [19] A. Burns and A. Wellings, *Real-Time Systems and Programming Languages*, 4th ed. Addison-Wesley, 2009.
- [20] S. K. Baruah, R. R. Howell, and L. E. Rosier, “Algorithms and complexity concerning the preemptive scheduling of periodic, real-time tasks on one processor,” *Real-Time Systems*, vol. 2, no. 4, pp. 301–324, 1990.
- [21] P. Lindgren, M. Lindner, A. Lindner *et al.*, “RTFM-core: Language and implementation of real-time reactive systems,” in *Proc. IEEE Int. Conf. Emerging Technologies and Factory Automation (ETFA)*, 2016, pp. 1–4.
- [22] S. Vestal, “Preemptive scheduling of multi-criticality systems with varying degrees of execution time assurance,” in *Proc. 28th IEEE Real-Time Systems Symposium (RTSS)*, 2007, pp. 239–243.
- [23] R. Wilhelm, J. Engblom, A. Ermedahl *et al.*, “The worst-case execution-time problem—overview of methods and survey of tools,” *ACM Trans. Embedded Comput. Syst.*, vol. 7, no. 3, pp. 36:1–36:53, 2008.
- [24] AbsInt GmbH, “aiT WCET Analyzer,” <https://www.absint.com/ait/>, 2024.
- [25] ARM Ltd., “Application note 321: ARM Cortex-M programming guide to memory barrier instructions,” 2013, aRM DAI 0321A.
- [26] F. J. Cazorla, E. Quiñones, T. Vardanega *et al.*, “PROARTIS: Probabilistically analyzable real-time systems,” *ACM Trans. Embedded Comput. Syst.*, vol. 12, no. 2s, pp. 94:1–94:26, 2013.
- [27] I. Griffin, J. Boylan, and P. Emberson, “Practical probabilistic WCET estimation using the coefficient of variation,” in *Proc. 34th Euromicro Conference on Real-Time Systems (ECRTS)*, 2022, pp. 7:1–7:22.
- [28] J. Abella, F. J. Cazorla, and E. Quiñones, “pWCET distribution fitting for measurement-based timing analysis: Method and toolchain,” *ACM Trans. Embedded Comput. Syst.*, vol. 22, no. 3, pp. 49:1–49:25, 2023.
- [29] G. Klein, K. Elphinstone, G. Heiser *et al.*, “seL4: Formal verification of an OS kernel,” in *Proc. 22nd ACM Symposium on Operating Systems Principles (SOSP)*, 2009, pp. 207–220.
- [30] R. Gu, Z. Shao, H. Chen *et al.*, “CertiKOS: An extensible architecture for building certified concurrent OS kernels,” in *Proc. 12th USENIX Symposium on Operating Systems Design and Implementation (OSDI)*, 2016, pp. 653–669.
- [31] A. Ferraiuolo, A. Baumann, C. Hawblitzel, and B. Parno, “Komodo: Using verification to disentangle secure-enclave hardware from software,” in *Proc. 26th ACM Symposium on Operating Systems Principles (SOSP)*, 2017, pp. 287–305.
- [32] International Electrotechnical Commission, “IEC 62304: Medical Device Software—Software Life Cycle Processes,” 2015.
- [33] A. VanHattum, D. Schwartz-Narbonne, N. Chong, and A. Sampson, “Verifying dynamic trait objects in Rust,” in *Proc. ACM/IEEE 44th Int. Conf. Software Engineering: Software Engineering in Practice (ICSE-SEIP)*, 2022, pp. 321–330.
- [34] G. Schalk, D. J. McFarland, T. Hinterberger, N. Birbaumer, and J. R. Wolpaw, “BCI2000: A general-purpose brain-computer interface (BCI) system,” *IEEE Trans. Biomed. Eng.*, vol. 51, no. 6, pp. 1034–1043, 2004.
- [35] A. Gramfort, M. Luessi, E. Larson *et al.*, “MEG and EEG data analysis with MNE-Python,” *Front. Neurosci.*, vol. 7, p. 267, 2013.
- [36] V. Jayaram and A. Barachant, “MOABB: Trustworthy algorithm benchmarking for BCIs,” *J. Neural Eng.*, vol. 15, no. 6, p. 066011, 2018.
- [37] M. Field, D. Stirling, Z. Pan, and F. Naghdy, “Closed-loop neurostimulation on FreeRTOS: Implementation and timing analysis,” *J. Neural Eng.*, vol. 16, no. 3, p. 036018, 2019.
- [38] M. Moakher, “A differential geometric approach to the geometric mean of symmetric positive-definite matrices,” *SIAM J. Matrix Anal. Appl.*, vol. 26, no. 3, pp. 735–747, 2005.
- [39] H. He and D. Wu, “Transfer learning for brain-computer interfaces: A Euclidean space data alignment approach,” *IEEE Trans. Biomed. Eng.*, vol. 67, no. 2, pp. 399–410, 2020.
- [40] UNESCO, “Recommendation on the ethics of neurotechnology,” UNESCO General Conference, 43rd Session, 2025.
- [41] STMicroelectronics, “RM0090: Reference manual STM32F405/415, STM32F407/417, STM32F427/437, STM32F429/439 advanced Arm-based 32-bit MCUs,” 2024, document DocID018909, Rev. 21, October 2024.
- [42] ARM Ltd., “ARMv7-M architecture reference manual,” 2021, document ARM DDI 0403E.e, Issue E.e, March 2021.
- [43] International Organization for Standardization, “ISO 14971: Medical devices—application of risk management to medical devices,” 2019.
- [44] H. Krawczyk and P. Eronen, “HMAC-based extract-and-expand key derivation function (HKDF),” IETF RFC 5869, 2010.
- [45] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd ed. Wiley-Interscience, 2006.
- [46] B. Blankertz, R. Tomioka, S. Lemm, M. Kawanabe, and K.-R. Müller, “Optimizing spatial filters for robust EEG single-trial analysis,” *IEEE Signal Process. Mag.*, vol. 25, no. 1, pp. 41–56, 2008.
- [47] J. R. Wolpaw, N. Birbaumer, W. J. Heetderks *et al.*, “Brain-computer interface technology: A review of the first international meeting,” *IEEE Trans. Rehabil. Eng.*, vol. 8, no. 2, pp. 164–173, 2000.
- [48] M. M. Madden, “Challenges using Linux as a real-time operating system,” *NASA Technical Memorandum*, no. TM-2020-220568, 2020.
- [49] P. Shenoy, M. Krauledat, B. Blankertz, R. P. N. Rao, and K.-R. Müller, “Towards adaptive classification for BCI,” *J. Neural Eng.*, vol. 3, no. 1, pp. R13–R23, 2006.
- [50] F. Lotte, L. Bougrain, A. Cichocki, M. Clerc, M. Congedo, A. Rakotomamonjy, and F. Yger, “A review of classification algorithms for EEG-based brain-computer interfaces: A 10 year update,” *J. Neural Eng.*, vol. 15, no. 3, p. 031005, 2018.